Unit-2

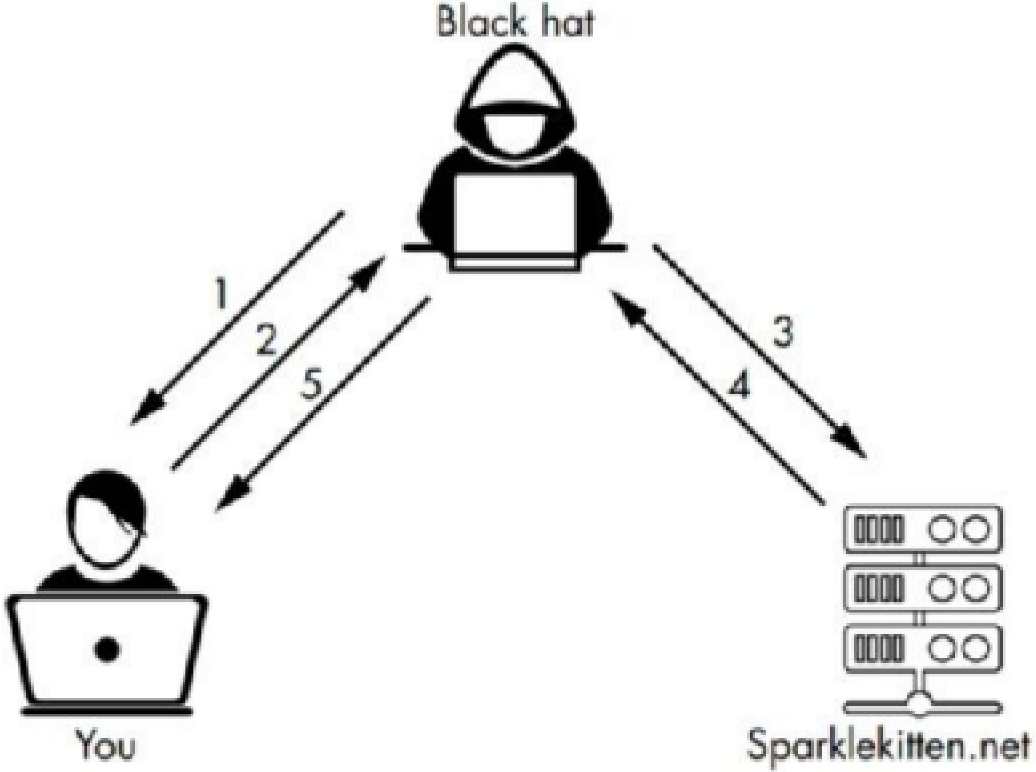
Ethical

Hacking

Reconnaissance: Introduction

- In cybersecurity, reconnaissance is the initial phase where attackers gather information about a target before launching an attack.
- This involves identifying potential vulnerabilities and weaknesses in a system, network, or organization.
- It's essentially the "research" phase for attackers, helping them understand their target's defenses and plan their attack strategy.

Reconnaissance: Introduction



Reconnaissance: Introduction

Types of Reconnaissance:

Passive Reconnaissance:

- This involves gathering information without directly interacting with the target system. Examples include:
- **Monitoring public channels:** Analyzing social media, public databases, and company websites for information.
- **Network traffic analysis:** Examining publicly available network traffic to identify patterns and potential vulnerabilities.

Types of Reconnaissance:

Active Reconnaissance:

● This involves directly interacting with the target system, often through scanning and probing. Examples include:
● Port scanning: Identifying open ports and services running on a system. ● Vulnerability scanning: Searching for known vulnerabilities in software and systems.

Types of Reconnaissance:

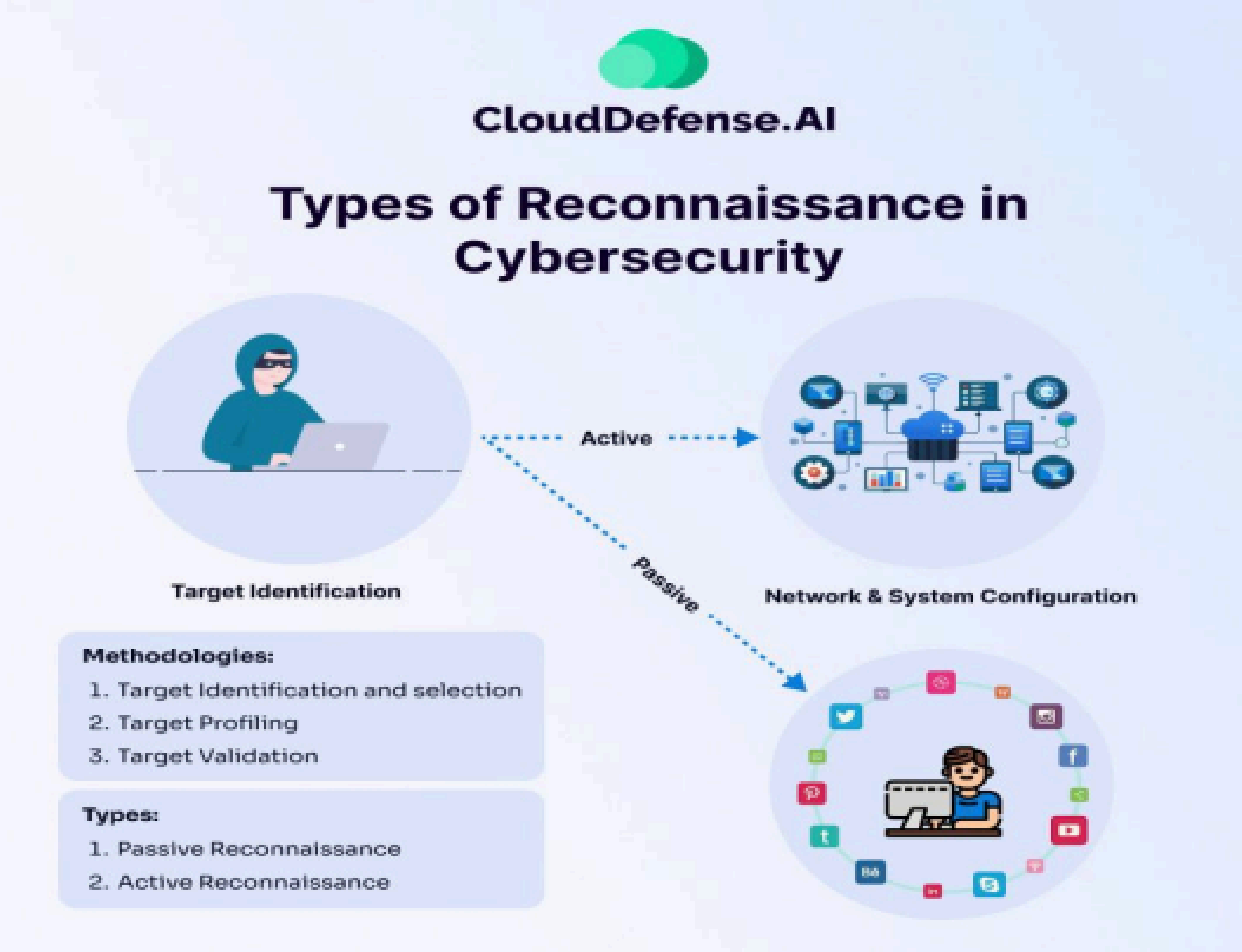
Social Reconnaissance:

- This involves gathering information about an organization's employees and culture, often through social engineering tactics.

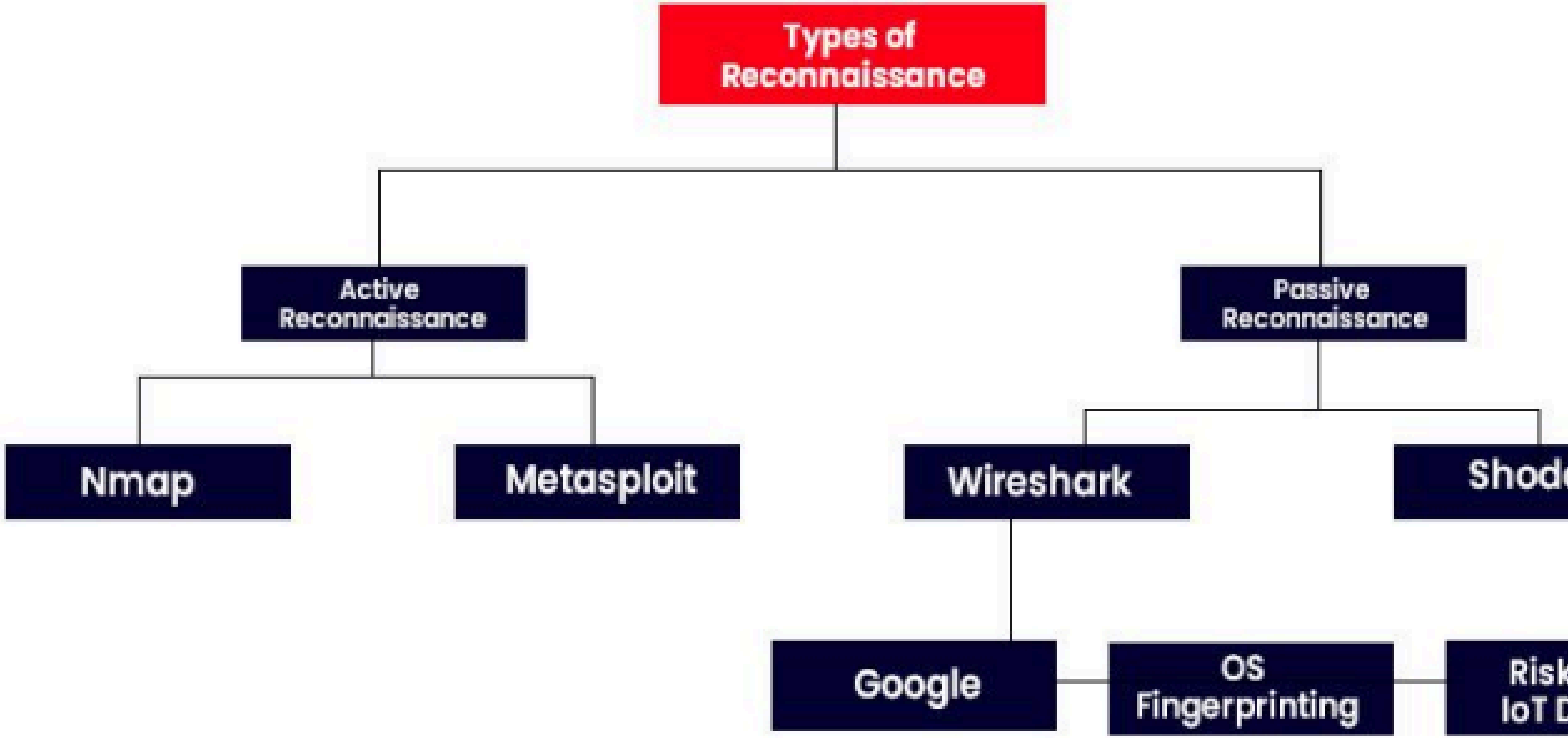
Types of Reconnaissance:

Importance of Reconnaissance:

- First phase of attacks: Reconnaissance is a crucial first step in most cyberattacks. ● Vulnerability identification: It helps attackers identify weaknesses and potential entry points.
- Strategic planning: Information gathered during reconnaissance allows attackers to tailor their attacks for maximum impact.


CloudDefense.AI
Types of Reconnaissance in Cybersecurity
Active
Passive
Target Identification
Network & System Configuration
Methodologies:
1. Target Identification and selection
2. Target Profiling
3. Target Validation
Types:
1. Passive Reconnaissance
2. Active Reconnaissance

Types of Reconnaissance: Tools available



Types of Reconnaissance
Active Reconnaissance
Passive Reconnaissance
Nmap
Metasploit
Wireshark
Google
OS Fingerprinting

Examples of Reconnaissance Techniques:

Footprinting:

Gathering basic information about a target, such as domain names, IP addresses, and contact information.

Scanning:

Using tools like Nmap to scan networks and identify open ports and services.

Enumeration:

Gathering more detailed information about a target, such as user accounts, network shares, and system configurations.

Social Engineering:

Using psychological manipulation to trick individuals into revealing sensitive information.

Detection and Prevention:

Monitor network traffic: Look for unusual traffic patterns, such as spikes in traffic to unfamiliar IP addresses.

Secure user accounts: Implement strong passwords and multi-factor authentication.

Train employees: Educate employees about social engineering tactics and phishing attacks.

Regularly update software: Patch vulnerabilities and keep systems up to date.

Implement security tools: Use firewalls, intrusion detection systems, and other security tools to detect and prevent attacks.